



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems
Academic Year 2024 – 2025(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS

Course Code & Title: AD3351 Design and Analysis of Algorithms

Name of the Faculty member: Ms.K.Usharani, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 4 / The Stable marriage Problem
- **Course Outcome:** CO 4 – Solve the problems using iterative improvement techniques for optimization.
- **Topic Learning Outcome:** 4a. Apply iterative improvement techniques for different real world problems and analyze its complexity
- **Activity Chosen:** Minute paper
- **Justification:**
 1. Minute paper activity will enable the students to recall the topics on their own.
 2. Better tool for brainstorming the ideas and concepts discussed in lecture.
 3. At the end of discussion on Stable marriage Problem, it is a better way to recall those concepts.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 1. The students were asked to take a paper and pen.
 2. They were asked to write the Stable marriage Problem and its algorithm individually.
 3. The students were asked to submit the paper.
 4. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO12	PSO2
CO5	3	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO12	PSO2
	3	1	1	1	2	1
Justification for correlation	Apply the knowledge of engineering fundamentals	Identify and analyze the iterative improvement concepts in complex engineering problems.	Design solutions for complex engineering problems by understanding the concepts of Stable marriage Problem	Function effectively as an individual by reflecting the ideas on the given topic.	Recognize the need and have the preparation and ability to engage in independent learning.	Motivates Continuous learning to provide the solution to real world business problems.

• **Images / Screenshot of the practice:**

Algorithm:

Input: A set of n men and a set of n women, along with rankings of the women by each man and rankings of the men by each woman with no ties allowed with the rankings.

Output: A Stable mate without other ties.

Step 1: The ranking matrix always starts with free men and free women.

Step 2: A man m and woman w .

def: of stable marriage problem: 953623244015 ✓

We consider an interesting version of bipartite matching called the stable marriage problem.

Consider a set $Y = \{m_1, m_2, \dots, m_n\}$ of n men and a set $X = \{w_1, w_2, \dots, w_n\}$ of n women.

Each man has a preference list order of women that a potential marriage partners with no ties allowed.

A marriage matching M is a set of n (m, w) pairs whose members are selected from disjoint n -element sets Y and X in a one-to-one fashion.

Each man m from Y is paired with exactly one woman w from X .

A pair (m, w) , where $m \in Y, w \in X$, is said to be a blocking pair.

blocking pair are the pairs which does not end up stable and makes matching unstable.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that they can able to recall the concepts of Stable marriage Problem. And this activity helped them to analyze their level of focus on the classroom.

- ❖ ***Benefit of the practice:***

1. Students were actively participated in this activity.
2. It increases students focus in class and also improves the student's writing skill and analyzing ability on the topic.

- ❖ ***Challenges faced in implementation:***

Some students felt difficult to recall the topics on their own.

References:

1. <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
2. <https://www.unl.edu/gradstudies/current/teaching/minute>
3. <https://oncourseworkshop.com/self-awareness/one-minute-paper/>